GRADE 7 ACCELERATED · UNIT 5

Multi-Step Proportional Relationships

Six lessons in teaching order · the next slide jumps to each one

Contents

5.01	Scale Drawings	slide 3
5.02–03	Scale Factor: Perimeter and Area	slide 38
5.04	Real-World Scale Drawings	slide 73
5.06–07	Converting Length Across Systems; Currency	slide 108
5.08–09	Converting Mass, Area and Volume	slide 143
5.10–11	Multi-Step Proportion Problems	slide 178

GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

Scale Drawings

MA.7.GR.1.5

I can find the scale factor between two figures, and use a scale to find a real length from a drawing or a drawing length from the real one.

Warm-Up

1. Find the value: $3 + 2 \times (-4)$
2. Find the value: $\sqrt{81}$
3. Write 2.5×10^3 as a number.
4. On a map, 3 cm stands for 12 km. What does 1 cm stand for?

Warm-Up

1. Find the value: $3 + 2 \times (-4)$ **–5**
2. Find the value: $\sqrt{81}$ **9**
3. Write 2.5×10^3 as a number. **2,500**
4. On a map, 3 cm stands for 12 km. What does 1 cm stand for?

   4 km

Vocabulary

scale drawing

a drawing of a real thing with every length multiplied by the same number: a map, a floor plan, a model

scale factor

the number every length is multiplied by, written k: from 4 cm to 6 cm, $k = 1.5$

corresponding sides

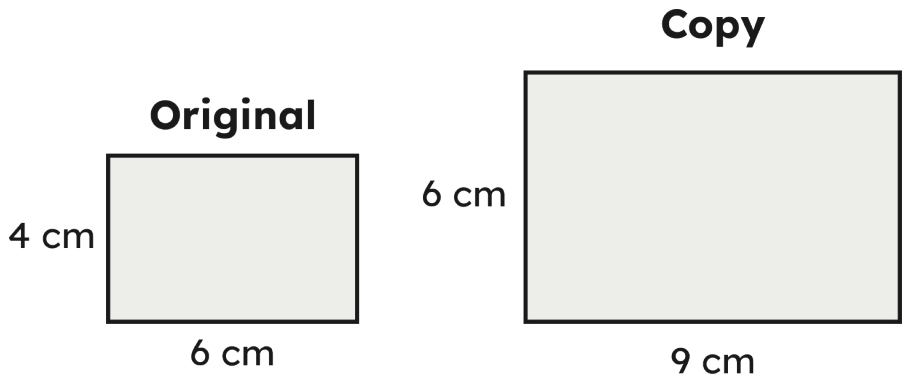
the side of the copy that matches a side of the original: the 6 cm side matches the 4 cm side

scale

a scale factor with units on it: 1 in : 20 mi means every inch on the map is 20 miles

Notes

I. A scale copy: every side times the same number



$$k = \frac{\text{copy}}{\text{original}} = \frac{9}{6} = 1.5 \qquad \frac{6}{4} = 1.5 \qquad \text{every side} \times 1.5$$

Every angle is still 90°: k multiplies lengths, not angles.

Notes

II. Find k: copy divided by original — and k can be less than 1

$$10 \text{ cm} \rightarrow 4 \text{ cm}: \quad k = \frac{4}{10} = 0.4 \quad \text{(not 2.5)}$$

k = copy ÷ original. Less than 1 shrinks, more than 1 enlarges, 1 is the same size.

Notes

III. Use k both ways

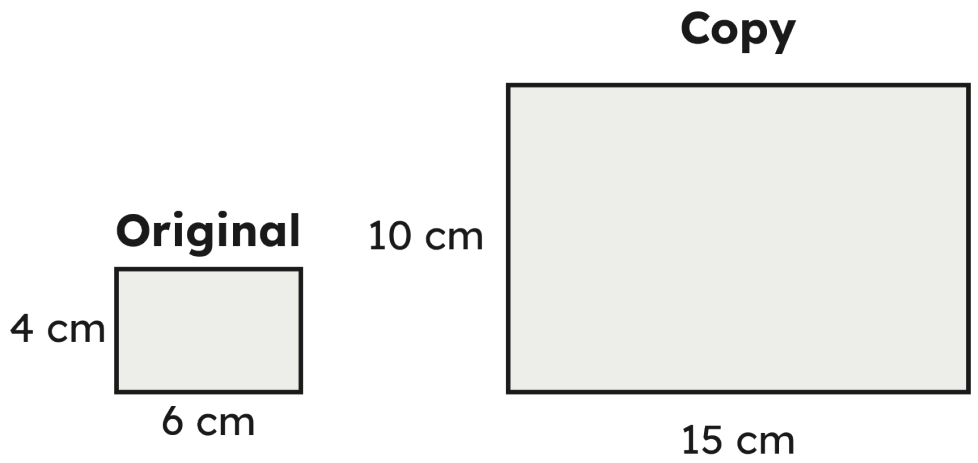
$$\text{copy} = \text{original} \times k \qquad \text{original} = \text{copy} \div k$$

$$1 \text{ in} : 20 \text{ mi} \quad 3.5 \text{ in} \times 20 = 70 \text{ mi} \quad 90 \text{ mi} \div 20 = 4.5 \text{ in}$$

To the copy, multiply by k. Back to the original, divide by k.

Example 1

A 6 cm by 4 cm rectangle is copied at 15 cm by 10 cm.



Find the scale factor.

Example 1

$$k = \frac{15}{6} = 2.5$$

$$\frac{10}{4} = 2.5$$

Answer: k = 2.5

Your Turn

An 8 cm by 10 cm rectangle is copied at 12 cm by 15 cm.

Find the scale factor.

Your Turn

An 8 cm by 10 cm rectangle is copied at 12 cm by 15 cm.

Find the scale factor.

$$k = \frac{12}{8} = 1.5$$

$$\frac{15}{10} = 1.5$$ — the same k

Answer: k = 1.5

Example 2

A map's scale is 1 in : 25 mi. Two towns are 2.6 in apart on the map.

How far apart are the towns?

Example 2

$2.6 \times 25 = 65$

Answer: 65 mi

Your Turn

A map's scale is 1 cm : 15 km. Two towns are 4.2 cm apart on the map.

How far apart are the towns?

Your Turn

A map's scale is 1 cm : 15 km. Two towns are 4.2 cm apart on the map.

How far apart are the towns?

$$4.2 \times 15 = 63$$

Answer: 63 km

Whiteboards · Question 1 of 9

A 2 cm side becomes 6 cm in the copy.

What is the scale factor?

Answer it.

Whiteboards · Question 1 of 9

A 2 cm side becomes 6 cm in the copy.

What is the scale factor?

$$k = \frac{6}{2} = 3$$

Answer: k = 3

Whiteboards · Question 2 of 9

An 8 cm side becomes 4 cm in the copy.

What is the scale factor?

Answer it.

Whiteboards · Question 2 of 9

An 8 cm side becomes 4 cm in the copy.

What is the scale factor?

$$k = \frac{4}{8} = \frac{1}{2}$$

Answer: k = ½

Whiteboards · Question 3 of 9

A 12 cm side becomes 9 cm in the copy.

What is the scale factor?

Answer it.

Whiteboards · Question 3 of 9

A 12 cm side becomes 9 cm in the copy.

What is the scale factor?

$$k = \frac{9}{12} = \frac{3}{4}$$

Answer: k = ¾

Whiteboards · Question 4 of 9

A side of 5 cm becomes 2 cm in the copy.

What is the scale factor?

A. 2.5

B. 0.4

C. 1

D. 3

Answer it.

Whiteboards · Question 4 of 9

A side of 5 cm becomes 2 cm in the copy.

What is the scale factor?

$$k = \frac{2}{5} = 0.4$$

Answer: B — 0.4

Whiteboards · Question 5 of 9

The scale factor is 3.5. A side of the original is 6 cm.

How long is that side in the copy?

Answer it.

Whiteboards · Question 5 of 9

The scale factor is 3.5. A side of the original is 6 cm.

How long is that side in the copy?

$$6 \times 3.5 = 21$$

Answer: 21 cm

Whiteboards · Question 6 of 9

A drawing has a 40° angle. The real figure is a scale copy at k = 2.5.

What is the angle in the real figure?

A. 40°

B. 42.5°

C. 16°

D. 100°

Answer it.

Whiteboards · Question 6 of 9

A drawing has a 40° angle. The real figure is a scale copy at k = 2.5.

What is the angle in the real figure?

Lengths are multiplied by k. Angles are not.

40°

Answer: A — 40°

Whiteboards · Question 7 of 9

A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14.

Which one is a scale copy of the first?

Answer it.

Whiteboards · Question 7 of 9

A triangle has sides 6, 8 and 10. Another has sides 9, 12 and 15; a third has sides 9, 12 and 14.

Which one is a scale copy of the first?

$\frac{9}{6} = \frac{12}{8} = \frac{15}{10} = 1.5$: the second is a copy

$\frac{14}{10} = 1.4 \neq 1.5$: the third is not

Answer: The 9-12-15 triangle (k = 1.5); the 9-12-14 one is not a copy

Whiteboards · Question 8 of 9

Dev, who is 13, draws a 15-ft wall as 3 in long.

How long is a 20-ft wall on his drawing?

Answer it.

Whiteboards · Question 8 of 9

Dev, who is 13, draws a 15-ft wall as 3 in long.

How long is a 20-ft wall on his drawing?

$15 \div 3 = 5$: every inch is 5 ft

$20 \div 5 = 4$

Answer: 4 in

Whiteboards · Question 9 of 9

Priya says the scale factor from a 6 cm side to its 3 cm copy is 2.

Is she right? Explain.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Priya says the scale factor from a 6 cm side to its 3 cm copy is 2.

Is she right? Explain.

$k = \frac{3}{6} = \frac{1}{2}$. No: the copy is smaller, so k is under 1.

Answer: No. k = 3 ÷ 6 = ½: the copy is smaller, so k is under 1.

Independent Set

1. A 5 cm side becomes 15 cm in the copy. Find the scale factor.
2. A 15 cm side becomes 6 cm in the copy. Find the scale factor.
3. A 3 cm by 5 cm rectangle is copied at 7.5 cm by 12.5 cm. Find the scale factor.
4. The scale factor is 1.5 and a side of the original is 8 cm. How long is that side in the copy?
5. A map's scale is 1 in : 25 mi. Two cities are 2.8 in apart on the map. How far apart are they?
6. On the same map, a road is 150 mi long. How long is it on the map?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Scale drawings: find the scale factor, missing side, or missing angle — WEA

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 2–3

Scale Factor: Perimeter and Area

MA.7.GR.1.5

I can find the perimeter and the area of a scale copy from the original and k, and find k from two areas.

Warm-Up

1. A 20 cm side becomes 8 cm in the copy. Find the scale factor.
2. Find the value: $\sqrt{144}$
3. Find the values: 3^2 and 3^3
4. Find the perimeter and the area of a 3 cm by 5 cm rectangle.

Warm-Up

1. A 20 cm side becomes 8 cm in the copy. Find the scale factor.

   k = 0.4

2. Find the value: $\sqrt{144}$ **12**

3. Find the values: 3^2 and 3^3 **9 and 27**

4. Find the perimeter and the area of a 3 cm by 5 cm rectangle.

   P = 16 cm, A = 15 cm²

Vocabulary

k squared

k^2, k times k: the number an area is multiplied by when every length is multiplied by k

perimeter ratio

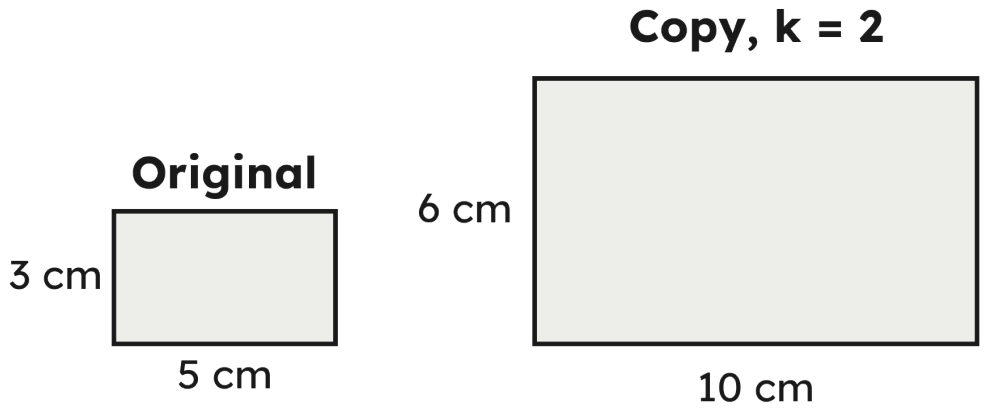
copy's perimeter ÷ original's perimeter — it equals k

area ratio

copy's area ÷ original's area — it equals k^2

Notes

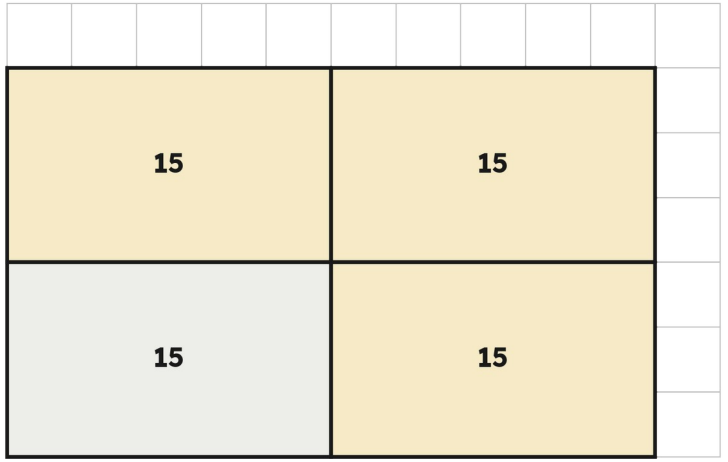
I. Every side × k, so the perimeter × k



$P : 16 \rightarrow 32$ $\quad$ $32 \div 16 = 2 = k$

Notes

II. Area is two lengths, so the area × k × k



the copy, 10 by 6, holds four of the original

$$A : 15 \rightarrow 60 \qquad 60 \div 15 = 4 = k^2$$

Notes

III. From two areas back to k: a square root

$$\frac{180}{20} = 9 = k^2 \qquad k = \sqrt{9} = 3 \qquad 4 \text{ cm} \times 3 = 12 \text{ cm}$$

Area ratio = k^2. k = the square root of the area ratio.

Example 1

A 2 cm by 4 cm rectangle is scaled by k = 3.

Find the copy's perimeter and area.

Example 1

$P = 2(2 + 4) = 12$ $\quad 12 \times 3 = 36$

$A = 2 \times 4 = 8$ $\quad 8 \times 3^2 = 72$

Answer: $P = 36$ cm, $A = 72$ cm²

Your Turn

A 3 cm by 6 cm rectangle is scaled by k = 4.

Find the copy's perimeter and area.

Your Turn

A 3 cm by 6 cm rectangle is scaled by k = 4.

Find the copy's perimeter and area.

$$P = 18 \times 4 = 72$$

$$A = 18 \times 4^2 = 18 \times 16 = 288$$

Answer: P = 72 cm, A = 288 cm²

Example 2

A figure has area 15 cm^2. Its scale copy has area 240 cm^2. A side of the original is 3 cm.

Find k, then the matching side of the copy.

Example 2

$$\frac{240}{15} = 16 = k^2$$

$$k = \sqrt{16} = 4 \qquad 3 \times 4 = 12$$

Answer: k = 4; the side is 12 cm

Your Turn

A figure has area 12 cm^2. Its scale copy has area 300 cm^2. A side of the original is 2 cm.

Find k, then the matching side of the copy.

Your Turn

A figure has area 12 cm². Its scale copy has area 300 cm². A side of the original is 2 cm.

Find k, then the matching side of the copy.

$$\frac{300}{12} = 25 = k^2$$

$$k = \sqrt{25} = 5 \qquad 2 \times 5 = 10$$

Answer: k = 5; the side is 10 cm

Whiteboards · Question 1 of 9

A figure with perimeter 14 cm is scaled by k = 2.

What is the copy's perimeter?

Answer it.

Whiteboards · Question 1 of 9

A figure with perimeter 14 cm is scaled by k = 2.

What is the copy's perimeter?

$$14 \times 2 = 28$$

Answer: 28 cm

Whiteboards · Question 2 of 9

A figure with area 12 cm^2 is scaled by k = 2.

What is the copy's area?

Answer it.

Whiteboards · Question 2 of 9

A figure with area 12 cm² is scaled by k = 2.

What is the copy's area?

$$12 \times 2^2 = 12 \times 4 = 48$$

Answer: 48 cm²

Whiteboards · Question 3 of 9

A figure with area 48 cm^2 is scaled by k = ½.

What is the copy's area?

Answer it.

Whiteboards · Question 3 of 9

A figure with area 48 cm² is scaled by k = ½.

What is the copy's area?

$$48 \times \left(\frac{1}{2}\right)^2 = 48 \times \frac{1}{4} = 12$$

Answer: 12 cm²

Whiteboards · Question 4 of 9

A figure with area 10 cm^2 is scaled by $k = 3$.

What is the copy's area?

A. 90

B. 60

C. 30

D. 13

Answer it.

Whiteboards · Question 4 of 9

A figure with area 10 cm² is scaled by k = 3.

What is the copy's area?

$$10 \times 3^2 = 10 \times 9 = 90$$

Answer: A — 90 cm²

Whiteboards · Question 5 of 9

A figure with area 8 cm^2 is scaled by k = 1.5.

What is the copy's area?

Answer it.

Whiteboards · Question 5 of 9

A figure with area 8 cm^2 is scaled by k = 1.5.

What is the copy's area?

$$1.5^2 = 2.25$$

$$8 \times 2.25 = 18$$

Answer: 18 cm^2

Whiteboards · Question 6 of 9

A figure's perimeter is 20 cm; its copy's perimeter is 60 cm.

What is the area ratio, copy to original?

A. 6

B. 3

C. 40

D. 9

Answer it.

Whiteboards · Question 6 of 9

A figure's perimeter is 20 cm; its copy's perimeter is 60 cm.

What is the area ratio, copy to original?

$k = \frac{60}{20} = 3$: the perimeter ratio is k

$k^2 = 9$: the area ratio

Answer: D — 9

Whiteboards · Question 7 of 9

A triangle with sides 6, 8 and 10 cm is scaled by k = ½. Its area is 24 cm^2.

Find the copy's perimeter and area.

Answer it.

Whiteboards · Question 7 of 9

A triangle with sides 6, 8 and 10 cm is scaled by k = ½. Its area is 24 cm².

Find the copy's perimeter and area.

$$P = 24 \times \frac{1}{2} = 12$$

$$A = 24 \times \frac{1}{4} = 6$$

Answer: P = 12 cm, A = 6 cm²

Whiteboards · Question 8 of 9

A 4 in by 6 in photo, taken 3 years ago, is enlarged at k = 2.5.

What is the enlargement's area?

Answer it.

Whiteboards · Question 8 of 9

A 4 in by 6 in photo, taken 3 years ago, is enlarged at k = 2.5.

What is the enlargement's area?

$$A = 4 \times 6 = 24$$

$$24 \times 2.5^2 = 24 \times 6.25 = 150$$

Answer: 150 in²

Whiteboards · Question 9 of 9

Jo says that doubling every side of a rectangle doubles its area.

Explain why not, with a counterexample.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Jo says that doubling every side of a rectangle doubles its area.

Explain why not, with a counterexample.

$$2 \times 7 = 14 \quad 4 \times 14 = 56 \quad 56 \div 14 = 4$$

Answer: No: 2 by 7 doubled is 4 by 14, area 14 → 56 — × 4, as both lengths doubled.

Independent Set

1. A figure with perimeter 18 cm is scaled by k = 3. Find the copy's perimeter.
2. A figure with area 7 cm^2 is scaled by k = 3. Find the copy's area.
3. A figure with area 40 cm^2 is scaled by k = ½. Find the copy's area.
4. A 2 cm by 5 cm rectangle is scaled by k = 2.5. Find the copy's perimeter and area.

5. A figure has area 9 cm^2 and its scale copy has area 144 cm^2. Find k.
6. A figure's perimeter is 12 cm and its copy's perimeter is 48 cm. What is the area ratio, copy to original?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Perimeter and area: changes in scale — ZC6

Scale drawings: find the scale factor or missing side — XHM

GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

Real-World Scale Drawings

MA.7.GR.1.5

I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

Warm-Up

1. A figure with area 9 cm^2 is scaled by k = 2. Find the copy's area.
2. A 5 cm side becomes 3 cm in the copy. Find the scale factor.
3. Find the product: 1.5×2.4
4. A car goes 90 miles in 2 hours. How far does it go per hour?

Warm-Up

1. A figure with area 9 cm^2 is scaled by k = 2. Find the copy's area.

   36 cm^2

2. A 5 cm side becomes 3 cm in the copy. Find the scale factor.

   k = 0.6

3. Find the product: 1.5×2.4 **3.6**

4. A car goes 90 miles in 2 hours. How far does it go per hour?

   45 miles per hour

Vocabulary

floor plan

a scale drawing of a room or building seen from above

scale (two units)

a ratio such as ¼ in : 1 ft — a drawing length and the real length it stands for

unit rate of a scale

the scale written per 1 drawing unit: ¼ in : 1 ft is 1 in : 4 ft

scale with one unit

1 : 50 — no units: 1 cm of model is 50 cm of car. 1 in : 4 ft is 1 : 48 this way, and 48 is the scale factor

Notes

I. A scale is a rate: write it per 1 unit

$$\frac{1}{4}\ \text{in} : 1\ \text{ft} \quad \rightarrow \quad 1\ \text{in} : 4\ \text{ft}$$

Drawing × rate = real. Real ÷ rate = drawing. With one unit, 1 in : 4 ft is 1 : 48.

Notes

II. Area: go to the real dimensions first

1 in : 4 ft



$$8 \text{ ft} \times 10 \text{ ft} = 80 \text{ ft}^2$$

Notes

III. Real to drawing is a division

$$14 \div 4 = 3.5 \text{ in}$$

No scale printed? One pair gives it: 2 in ↔ 9 ft is 4.5 ft per inch.

Example 1

A floor plan is drawn at ¼ in : 1 ft. A room is 2.5 in by 3 in on the plan.

Find the room's real dimensions and area.

Example 1

$\frac{1}{4}$ in : 1 ft = 1 in : 4 ft

$2.5 \times 4 = 10$ $\quad$ $3 \times 4 = 12$

$10 \times 12 = 120$

Answer: 10 ft by 12 ft; 120 ft²

Your Turn

A floor plan is drawn at ¼ in : 1 ft. A room is 1.5 in by 4 in on the plan.

Find the room's real dimensions and area.

Your Turn

A floor plan is drawn at ¼ in : 1 ft. A room is 1.5 in by 4 in on the plan.

Find the room's real dimensions and area.

$1.5 \times 4 = 6$ $\quad$ $4 \times 4 = 16$

$6 \times 16 = 96$

Answer: 6 ft by 16 ft; 96 ft²

Example 2

On a drawing a wall is 4 in long; the real wall is 18 ft. Another wall is 6 in long on the drawing.

How long is the second wall really?

Example 2

$18 \div 4 = 4.5$ ft per in

$6 \times 4.5 = 27$

Answer: 27 ft

Your Turn

On a drawing a wall is 2 in long; the real wall is 13 ft. Another wall is 5 in long on the drawing.

How long is the second wall really?

Your Turn

On a drawing a wall is 2 in long; the real wall is 13 ft. Another wall is 5 in long on the drawing.

How long is the second wall really?

$$13 \div 2 = 6.5 \text{ ft per in}$$

$$5 \times 6.5 = 32.5$$

Answer: 32.5 ft

Whiteboards · Question 1 of 9

A plan is drawn at ½ in : 1 ft. A wall is 4 in long on it.

How long is the real wall?

Answer it.

Whiteboards · Question 1 of 9

A plan is drawn at ½ in : 1 ft. A wall is 4 in long on it.

How long is the real wall?

$$\frac{1}{2} \text{ in} : 1 \text{ ft} = 1 \text{ in} : 2 \text{ ft}$$

$$4 \times 2 = 8$$

Answer: 8 ft

Whiteboards · Question 2 of 9

A plan is drawn at 1 in : 3 ft. A room is 2.5 in long on it.

How long is the real room?

Answer it.

Whiteboards · Question 2 of 9

A plan is drawn at 1 in : 3 ft. A room is 2.5 in long on it.

How long is the real room?

$$2.5 \times 3 = 7.5$$

Answer: 7.5 ft

Whiteboards · Question 3 of 9

A plan is drawn at 1 in : 3 ft. A real wall is 21 ft long.

How long is it on the plan?

Answer it.

Whiteboards · Question 3 of 9

A plan is drawn at 1 in : 3 ft. A real wall is 21 ft long.

How long is it on the plan?

$$21 \div 3 = 7$$

Answer: 7 in

Whiteboards · Question 4 of 9

A model car is built at 1 : 50 and is 8 cm long.

How long is the real car?

A. 4 m

B. 0.16 m

C. 400 m

D. 58 cm

Answer it.

Whiteboards · Question 4 of 9

A model car is built at 1 : 50 and is 8 cm long.

How long is the real car?

$$8 \times 50 = 400 \text{ cm}$$

$$400 \div 100 = 4 \text{ m}$$

Answer: A — 4 m

Whiteboards · Question 5 of 9

A plan is drawn at ¼ in : 1 ft. A real wall is 15 ft long.

How long is it on the plan?

Answer it.

Whiteboards · Question 5 of 9

A plan is drawn at ¼ in : 1 ft. A real wall is 15 ft long.

How long is it on the plan?

$1 \text{ in} : 4 \text{ ft} \qquad 15 \div 4 = 3.75$

Answer: 3.75 in

Whiteboards · Question 6 of 9

A plan is drawn at 1 in : 4 ft. A room is 2 in by 3 in on it.

What is the room's real area?

A. $24\ \text{ft}^2$

B. $96\ \text{ft}^2$

C. $40\ \text{ft}^2$

D. $20\ \text{ft}^2$

Answer it.

Whiteboards · Question 6 of 9

A plan is drawn at 1 in : 4 ft. A room is 2 in by 3 in on it.

What is the room's real area?

$$2 \times 4 = 8 \qquad 3 \times 4 = 12$$

$$8 \times 12 = 96$$

Answer: B — 96 ft²

Whiteboards · Question 7 of 9

A plane is 36 ft long. One model of it is built at 1 : 48, another at 1 : 72.

Which model is longer, and by how much?

Answer it.

Whiteboards · Question 7 of 9

A plane is 36 ft long. One model of it is built at 1 : 48, another at 1 : 72.

Which model is longer, and by how much?

$$36 \times 12 = 432 \text{ in}$$

$$432 \div 48 = 9 \qquad 432 \div 72 = 6$$

Answer: The 1 : 48 model, by 3 in (9 in against 6 in)

Whiteboards · Question 8 of 9

Mia, who is 13, enlarges her 9 in by 12 in poster into a mural at 1 in : 2 ft.

What are the mural's dimensions?

Answer it.

Whiteboards · Question 8 of 9

Mia, who is 13, enlarges her 9 in by 12 in poster into a mural at 1 in : 2 ft.

What are the mural's dimensions?

$$9 \times 2 = 18 \qquad 12 \times 2 = 24$$

Answer: 18 ft by 24 ft

Whiteboards · Question 9 of 9

On Andrew's drawing a wall is 3.5 in wide and 4 in tall; the real wall is 26.25 ft wide.

Find the real height and the real area. Which step uses the scale?

Write your answer in sentences.

Whiteboards · Question 9 of 9

On Andrew's drawing a wall is 3.5 in wide and 4 in tall; the real wall is 26.25 ft wide.

Find the real height and the real area. Which step uses the scale?

$26.25 \div 3.5 = 7.5$ ft per in $\qquad 4 \times 7.5 = 30$

$26.25 \times 30 = 787.5$

Answer: 30 ft high; 787.5 ft². The scale (7.5 ft per in) is the step from 4 in to 30 ft.

Independent Set

1. A plan is drawn at ½ in : 1 ft. A wall is 6 in long on it. Find the real length.

2. A plan is drawn at 1 in : 5 ft. A room is 3.2 in long on it. Find the real length.

3. A plan is drawn at 1 in : 5 ft. A real wall is 22 ft long. Find its length on the plan.

4. A plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on it. Find the real dimensions and area.

5. A model boat is built at 1 : 40 and is 15 cm long. Find the real boat's length in meters.

6. On a drawing a wall is 3 in long; the real wall is 16.5 ft. Another wall is 5 in long on the drawing. How long is it really?

IXL

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2. Work on paper where the question needs work. The answer box does not show it.
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Today's skills — all required

Scale drawings: word problems — 84H

Scale drawings: scale factor word problems — KCM

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 6–7

Converting Length Across Systems; Currency

MA.7.AR.3.3

I can convert a length between the customary and metric systems with a ratio from the reference sheet, and convert money with a given rate.

Warm-Up

1. At ¼ in : 1 ft, how long is a 10-ft wall on the plan?
2. A 14 cm side becomes 7 cm in the copy. Find the scale factor.
3. Find the quotient: $4.5 \div 0.9$
4. 12 in = 1 ft. How many feet is 30 in?

Warm-Up

1. At ¼ in : 1 ft, how long is a 10-ft wall on the plan? **2.5 in**
2. A 14 cm side becomes 7 cm in the copy. Find the scale factor.

   k = ½
3. Find the quotient: $4.5 \div 0.9$ **5**
4. 12 in = 1 ft. How many feet is 30 in? **2.5 ft**

Vocabulary

customary

the inch–foot–mile–pound–gallon system

metric

the centimeter–meter–kilometer–kilogram–liter system

conversion ratio

a pair of equal measures, such as 1 in = 2.54 cm, written as a fraction that equals 1

exchange rate

how much of one currency one unit of another buys — a rate someone sets, printed on the problem

Notes

I. A conversion ratio is a form of 1

$$10 \text{ in} \times \frac{2.54 \text{ cm}}{1 \text{ in}} = 25.4 \text{ cm}$$

Notes

II. Which ratio? Estimate first

$$50 \text{ cm} \times \frac{0.39 \text{ in}}{1 \text{ cm}} = 19.5 \text{ in}$$

Pick the ratio whose 1 is the unit you have. Estimate: which is the bigger unit?

Notes

III. Money is a rate someone sets

$$1 \text{ dollar} = 0.92 \text{ euros}$$

$$150 \times 0.92 = 138 \text{ euros} \qquad 92 \div 0.92 = 100$$

Example 1

1 in = 2.54 cm. A tablet screen is 15 in across.

How many centimeters is that?

Example 1

$$15 \times 2.54 = 38.1$$

Answer: 38.1 cm

Your Turn

1 in = 2.54 cm. A monitor is 25 in across.

How many centimeters is that?

Your Turn

1 in = 2.54 cm. A monitor is 25 in across.

How many centimeters is that?

$$25 \times 2.54 = 63.5$$

Answer: 63.5 cm

Example 2

Today's rate: 1 US dollar = 0.92 euros.

How many euros is $80? And how many dollars is 46 euros?

Example 2

$80 \times 0.92 = 73.60$

$46 \div 0.92 = 50$

Answer: 73.60 euros; $50

Your Turn

Today's rate: 1 US dollar = 1.35 Canadian dollars (CAD).

$120 in CAD? And 81 CAD in US dollars?

Your Turn

Today's rate: 1 US dollar = 1.35 Canadian dollars (CAD).

$120 in CAD? And 81 CAD in US dollars?

$$120 \times 1.35 = 162$$

$$81 \div 1.35 = 60$$

Answer: 162 CAD; $60

Whiteboards · Question 1 of 9

1 in = 2.54 cm. A ruler is 20 in long.

How many centimeters is that?

Answer it.

Whiteboards · Question 1 of 9

1 in = 2.54 cm. A ruler is 20 in long.

How many centimeters is that?

$$20 \times 2.54 = 50.8$$

Answer: 50.8 cm

Whiteboards · Question 2 of 9

1 cm = 0.39 in. A phone is 20 cm long.

How many inches is that?

Answer it.

Whiteboards · Question 2 of 9

1 cm = 0.39 in. A phone is 20 cm long.

How many inches is that?

$$20 \times 0.39 = 7.8$$

Answer: 7.8 in

Whiteboards · Question 3 of 9

1 mi = 1.61 km. A race is 5 mi long.

How many kilometers is that?

Answer it.

Whiteboards · Question 3 of 9

1 mi = 1.61 km. A race is 5 mi long.

How many kilometers is that?

$$5 \times 1.61 = 8.05$$

Answer: 8.05 km

Whiteboards · Question 4 of 9

1 km = 0.62 mi. A trail is 5 km long.

How many miles is that?

A. 5.62 mi

B. 0.31 mi

C. 8.05 mi

D. 3.1 mi

Answer it.

Whiteboards · Question 4 of 9

1 km = 0.62 mi. A trail is 5 km long.

How many miles is that?

$$5 \times 0.62 = 3.1$$

Answer: D — 3.1 mi

Whiteboards · Question 5 of 9

1 ft = 12 in and 1 in = 2.54 cm. A door is 6 ft tall.

How many centimeters is that?

Answer it.

Whiteboards · Question 5 of 9

1 ft = 12 in and 1 in = 2.54 cm. A door is 6 ft tall.

How many centimeters is that?

$$6 \times 12 = 72 \text{ in}$$

$$72 \times 2.54 = 182.88$$

Answer: 182.88 cm

Whiteboards · Question 6 of 9

1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. A table is 3 yd long.

About how many centimeters is that? Estimate, then compute.

Answer it.

Whiteboards · Question 6 of 9

1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. A table is 3 yd long.

About how many centimeters is that? Estimate, then compute.

Estimate: 3 yd is a bit under 3 m, so about 270 cm

$3 \times 3 \times 12 = 108$ in $108 \times 2.54 = 274.32$

Answer: About 270; exactly 274.32 cm

Whiteboards · Question 7 of 9

Rate on Nov 2: $1 = 1.32 CAD; 1 CAD = 16.2 pesos. Theo changes $200 to CAD, then to pesos.

How many pesos does he get?

Answer it.

Whiteboards · Question 7 of 9

Rate on Nov 2: $1 = 1.32 CAD; 1 CAD = 16.2 pesos. Theo changes $200 to CAD, then to pesos.

How many pesos does he get?

$$200 \times 1.32 = 264 \qquad 264 \times 16.2 = 4{,}276.80$$

Answer: 4,276.80 pesos

Whiteboards · Question 8 of 9

1 km = 0.62 mi. Theo, who weighs 110 lb, rides a 12 km trail.

How long is the trail in miles?

Answer it.

Whiteboards · Question 8 of 9

1 km = 0.62 mi. Theo, who weighs 110 lb, rides a 12 km trail.

How long is the trail in miles?

$$12 \times 0.62 = 7.44$$

Answer: 7.44 mi

Whiteboards · Question 9 of 9

1 in = 2.54 cm; 1 cm = 0.39 in. Jo converts 30 cm to inches: 30 × 2.54 = 76.2 in.

Name the error and fix it.

Write your answer in sentences.

Whiteboards · Question 9 of 9

1 in = 2.54 cm; 1 cm = 0.39 in. Jo converts 30 cm to inches: 30 × 2.54 = 76.2 in.

Name the error and fix it.

$$30 \times 0.39 = 11.7$$

Answer: She used the inches ratio on centimeters. Inches are bigger, so fewer: 11.7 in.

Independent Set

1. 1 in = 2.54 cm. How many centimeters is 12 in?
2. 1 cm = 0.39 in. How many inches is 40 cm?
3. 1 mi = 1.61 km. How many kilometers is 10 mi?
4. 1 km = 0.62 mi. How many miles is 8 km?
5. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 2 yd? Estimate first.
6. Today's rate: 1 US dollar = 0.92 euros. How many euros is $50? How many dollars is 23 euros?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between customary and metric systems — E8Z

Compare and convert customary units — 4B8

Compare and convert metric units — HW2

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 8–9

Converting Mass, Area and Volume

MA.7.AR.3.3

I can convert mass, volume and area between systems, and square the length ratio for area.

Warm-Up

1. 1 in = 2.54 cm. How many centimeters is 12 in?
2. A figure with area 7 cm^2 is scaled by k = 3. Find the copy's area.
3. Find the product: -2.5×4
4. 1 ft = 12 in. A 1-ft square has what area in square inches?

Warm-Up

1. 1 in = 2.54 cm. How many centimeters is 12 in? **30.48 cm**
2. A figure with area 7 cm^2 is scaled by k = 3. Find the copy's area.

   63 cm^2
3. Find the product: -2.5×4 **–10**
4. 1 ft = 12 in. A 1-ft square has what area in square inches?

   144 in^2

Vocabulary

mass and weight

pounds and kilograms — one word on this test

ounce (oz)

a weight: 1 lb = 16 oz, 1 oz = 28.35 g

fluid ounce (fl oz)

a volume: 1 c = 8 fl oz — not the same unit as the ounce

square units

ft^2, cm^2, m^2 — an area; the length ratio is used twice

cubic units

ft^3, in^3 — a volume; the length ratio is used three times

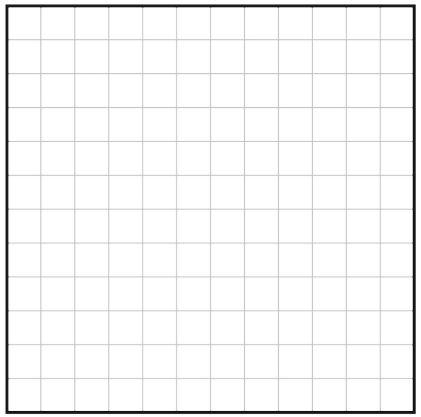
Notes

I. Mass and volume convert like length

$$50 \text{ lb} \times 0.454 = 22.7 \text{ kg} \qquad 2 \text{ gal} \times 3.785 = 7.57 \text{ L}$$

Notes

II. Area: the length ratio is used twice



1 ft² is 12 in by 12 in: 144 in²

$$1\ \text{ft}^2 = 12 \times 12 = 144\ \text{in}^2 \qquad 3\ \text{ft}^2 = 432\ \text{in}^2$$

$$1\ \text{ft} = 0.305\ \text{m}, \text{ so } 1\ \text{ft}^2 = 0.305^2 \approx 0.093\ \text{m}^2$$

Notes

III. Fluid ounces go through cups

$$12 \text{ fl oz} \div 8 = 1.5 \text{ c} \qquad 1.5 \times 0.24 = 0.36 \text{ L}$$

Length: the ratio once. Area: twice. Volume: three times.

Example 1

1 lb = 0.454 kg. A suitcase weighs 70 lb.

How many kilograms is that?

Example 1

$$70 \times 0.454 = 31.78$$

Answer: 31.78 kg

Your Turn

1 lb = 0.454 kg. A crate weighs 90 lb.

How many kilograms is that?

Your Turn

1 lb = 0.454 kg. A crate weighs 90 lb.

How many kilograms is that?

$$90 \times 0.454 = 40.86$$

Answer: 40.86 kg (about 40.9)

Example 2

1 ft = 12 in. A tile is 5 ft^2 in area.

How many square inches is that?

Example 2

$1\ ft^2 = 12^2 = 144\ in^2$

$5 \times 144 = 720$

Answer: 720 in^2

Your Turn

1 yd = 3 ft. A patio is 5 yd^2 in area.

How many square feet is that?

Your Turn

1 yd = 3 ft. A patio is 5 yd^2 in area.

How many square feet is that?

$3^2 = 9$ ft^2 in a square yard

$5 \times 9 = 45$

Answer: 45 ft^2

Whiteboards · Question 1 of 9

1 kg = 2.204 lb. A bag of rice is 10 kg.

How many pounds is that?

Answer it.

Whiteboards · Question 1 of 9

1 kg = 2.204 lb. A bag of rice is 10 kg.

How many pounds is that?

$$10 \times 2.204 = 22.04$$

Answer: 22.04 lb

Whiteboards · Question 2 of 9

1 gal = 3.785 L. A cooler holds 4 gal.

How many liters is that?

Answer it.

Whiteboards · Question 2 of 9

1 gal = 3.785 L. A cooler holds 4 gal.

How many liters is that?

$$4 \times 3.785 = 15.14$$

Answer: 15.14 L

Whiteboards · Question 3 of 9

1 L = 0.264 gal. A tank holds 10 L.

How many gallons is that?

Answer it.

Whiteboards · Question 3 of 9

1 L = 0.264 gal. A tank holds 10 L.

How many gallons is that?

$$10 \times 0.264 = 2.64$$

Answer: 2.64 gal

Whiteboards · Question 4 of 9

1 yd = 3 ft. A patch of lawn is 2 yd^2 in area.

How many square feet is that?

A. 12 ft^2

B. 18 ft^2

C. 9 ft^2

D. 6 ft^2

Answer it.

Whiteboards · Question 4 of 9

1 yd = 3 ft. A patch of lawn is 2 yd^2 in area.

How many square feet is that?

$3^2 = 9$ ft^2 in a square yard

$2 \times 9 = 18$

Answer: B — 18 ft^2

Whiteboards · Question 5 of 9

1 c = 8 fl oz and 1 c = 0.24 L. A bottle holds 16 fl oz.

How many liters is that?

Answer it.

Whiteboards · Question 5 of 9

1 c = 8 fl oz and 1 c = 0.24 L. A bottle holds 16 fl oz.

How many liters is that?

$$16 \div 8 = 2 \text{ c}$$

$$2 \times 0.24 = 0.48$$

Answer: 0.48 L

Whiteboards · Question 6 of 9

1 m = 100 cm.

How many square centimeters is 1 m^2?

A. 1,000

B. 200

C. 10,000

D. 100

Answer it.

Whiteboards · Question 6 of 9

1 m = 100 cm.

How many square centimeters is 1 m²?

$$100 \times 100 = 10{,}000$$

Answer: C — 10,000 cm²

Whiteboards · Question 7 of 9

1 kg = 2.204 lb. A 5-lb bag of flour and a 2.5-kg bag of sugar.

Which is heavier, and by how much?

Answer it.

Whiteboards · Question 7 of 9

1 kg = 2.204 lb. A 5-lb bag of flour and a 2.5-kg bag of sugar.

Which is heavier, and by how much?

$$2.5 \times 2.204 = 5.51 \text{ lb}$$

$$5.51 - 5 = 0.51$$

Answer: The sugar, by 0.51 lb

Whiteboards · Question 8 of 9

1 gal = 3.785 L. A 20-gal fish tank, bought for $45, is being filled.

How many liters does it hold?

Answer it.

Whiteboards · Question 8 of 9

1 gal = 3.785 L. A 20-gal fish tank, bought for $45, is being filled.

How many liters does it hold?

$$20 \times 3.785 = 75.7$$

Answer: 75.7 L

Whiteboards · Question 9 of 9

1 ft = 12 in. Priya says 2 ft^3 is 24 in^3, because 2 × 12 = 24.

Name the error and fix it.

Write your answer in sentences.

Whiteboards · Question 9 of 9

1 ft = 12 in. Priya says 2 ft^3 is 24 in^3, because 2 × 12 = 24.

Name the error and fix it.

$12^3 = 1{,}728$ $\qquad$ $2 \times 1{,}728 = 3{,}456$

Answer: She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in^3, so 2 ft^3 = 3,456 in^3.

Independent Set

1. 1 lb = 0.454 kg. How many kilograms is 30 lb?
2. 1 kg = 2.204 lb. How many pounds is 25 kg?
3. 1 gal = 3.785 L. How many liters is 6 gal?
4. 1 ft = 12 in. How many square inches is 9 ft^2?
5. 1 c = 8 fl oz and 1 c = 0.24 L. How many liters is 24 fl oz?
6. 1 ft = 0.305 m. How many square meters is 30 ft^2? (Round to two places.)

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between customary and metric systems — E8Z

Compare and convert customary units — 4B8

Compare and convert metric units — HW2

GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

Multi-Step Proportion Problems

MA.7.AR.4.5

I can solve a problem that takes two or more proportional steps, including a unit conversion, and compare two rates given in different units.

Warm-Up

1. 1 kg = 2.204 lb. How many pounds is 4 kg?
2. At 1 cm : 50 km, 3 cm on the map is how many km?
3. Find the value: $3 \times 2.5 + 1.5$
4. A car goes 180 miles in 3 hours. Its speed in miles per hour?

Warm-Up

1. 1 kg = 2.204 lb. How many pounds is 4 kg? **8.816 lb**
2. At 1 cm : 50 km, 3 cm on the map is how many km? **150 km**
3. Find the value: $3 \times 2.5 + 1.5$ **9**
4. A car goes 180 miles in 3 hours. Its speed in miles per hour?

   60 miles per hour

Vocabulary

unit rate

how much of one quantity per 1 of the other — 23 miles per 1 gallon

proportional

two quantities whose ratio stays the same: double one, the other doubles

per

for each 1 — the word that marks a unit rate

Notes

I. One unit rate, used twice

$23 \times 17.5 = 402.5$ mi per tank $1{,}207 \div 402.5 \approx 3$ tanks

Notes

II. Compare rates in one unit

$$100 \text{ km/h} \times 0.62 = 62 \text{ mi/h} \qquad 62 < 65$$

Same unit first. Then compare.

Notes

III. Scale a recipe, then check the pantry

$$k = \frac{30}{12} = 2.5 \qquad 2.5 \times 1\frac{1}{2} = 3\frac{3}{4} \text{ cups} > 3$$

Paint $1:2:3$, 4 gal of the first: $k = 4$, $6 \text{ parts} \times 4 = 24$ gal

Example 1

1 mi = 1.61 km. Runner A does 5 mi in 40 min; runner B does 9 km in 45 min.

Find each speed in km/h. Who is faster?

Example 1

$5 \times 1.61 = 8.05$ km 40 min $= \frac{2}{3}$ h

$8.05 \div \frac{2}{3} = 12.075$

$9 \div \frac{3}{4} = 12$

$12.075 > 12$

Answer: A ≈ 12.08 km/h, B = 12 km/h: A

Your Turn

1 mi = 1.61 km. Runner A does 4 mi in 30 min; runner B does 7 km in 35 min.

Find each speed in km/h. Who is faster?

Your Turn

1 mi = 1.61 km. Runner A does 4 mi in 30 min; runner B does 7 km in 35 min.

Find each speed in km/h. Who is faster?

$$4 \times 1.61 = 6.44 \qquad 6.44 \div \frac{1}{2} = 12.88$$

$$7 \div \frac{7}{12} = 12$$

Answer: A = 12.88 km/h, B = 12 km/h: A

Example 2

A recipe for 8 servings uses 2½ cups of rice.

How much rice for 20 servings? With 5 cups on hand, how many servings can be made?

Example 2

$$k = \frac{20}{8} = 2.5 \qquad 2.5 \times 2\frac{1}{2} = 6\frac{1}{4}$$

$$5 \div 2\frac{1}{2} = 2 \qquad 2 \times 8 = 16$$

Answer: 6¼ cups; 16 servings

Your Turn

A recipe for 6 servings uses 1½ cups of rice.

How much rice for 12 servings? With 4½ cups on hand, how many servings?

Your Turn

A recipe for 6 servings uses 1½ cups of rice.

How much rice for 12 servings? With 4½ cups on hand, how many servings?

$$k = \frac{12}{6} = 2 \qquad 2 \times 1\frac{1}{2} = 3$$

$$4\frac{1}{2} \div 1\frac{1}{2} = 3 \qquad 3 \times 6 = 18$$

Answer: 3 cups; 18 servings

Whiteboards · Question 1 of 9

A car gets 28 miles per gallon. Its tank holds 15 gal.

How far does it go on a full tank?

Answer it.

Whiteboards · Question 1 of 9

A car gets 28 miles per gallon. Its tank holds 15 gal.

How far does it go on a full tank?

$$28 \times 15 = 420$$

Answer: 420 mi

Whiteboards · Question 2 of 9

A printer prints 35 pages per minute. A job is 1,400 pages.

How long does the job take?

Answer it.

Whiteboards · Question 2 of 9

A printer prints 35 pages per minute. A job is 1,400 pages.

How long does the job take?

$$1{,}400 \div 35 = 40$$

Answer: 40 min

Whiteboards · Question 3 of 9

Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days.

How much does she earn?

Answer it.

Whiteboards · Question 3 of 9

Mia earns $14.50 per hour at a shop 2 mi from home, 6 h a day for 5 days.

How much does she earn?

$$6 \times 5 = 30 \qquad 30 \times 14.50 = 435$$

Answer: $435

Whiteboards · Question 4 of 9

A car gets 30 miles per gallon with a 12-gal tank. It makes a 900-mi trip.

How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)

A. 2

B. 75

C. 25

D. 3

Answer it.

Whiteboards · Question 4 of 9

A car gets 30 miles per gallon with a 12-gal tank. It makes a 900-mi trip.

How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.)

$$30 \times 12 = 360 \text{ mi per tank}$$

$$900 \div 360 = 2.5 \rightarrow 3 \text{ tanks}$$

Answer: D — 3 tanks

Whiteboards · Question 5 of 9

A cereal has 12 g of protein per 100 g.

How many grams of protein in 1 kg?

Answer it.

Whiteboards · Question 5 of 9

A cereal has 12 g of protein per 100 g.

How many grams of protein in 1 kg?

$$1{,}000 \div 100 = 10 \qquad 10 \times 12 = 120$$

Answer: 120 g

Whiteboards · Question 6 of 9

1 mi = 1.61 km. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min.

Who is faster?

A. Tie: 12 and 12

B. A: 12 > 7

C. B: 15.0 > 14.4

D. A: 14.4 > 11.27

Answer it.

Whiteboards · Question 6 of 9

1 mi = 1.61 km. Cyclist A rides 12 km in 50 min; cyclist B rides 7 mi in 45 min.

Who is faster?

$$\text{A}: 50 \text{ min} = \frac{5}{6} \text{ h} \qquad 12 \div \frac{5}{6} = 14.4 \text{ km/h}$$

$$\text{B}: 7 \times 1.61 = 11.27 \text{ km} \qquad 11.27 \div \frac{3}{4} \approx 15.03 \text{ km/h}$$

Answer: C — cyclist B

Whiteboards · Question 7 of 9

Paint is mixed white : blue : grey = 2 : 4 : 5. A batch uses 8 gal of white.

How many gallons is the whole batch?

Answer it.

Whiteboards · Question 7 of 9

Paint is mixed white : blue : grey = 2 : 4 : 5. A batch uses 8 gal of white.

How many gallons is the whole batch?

$$k = \frac{8}{2} = 4 \qquad 2 + 4 + 5 = 11 \text{ parts}$$

$$11 \times 4 = 44$$

Answer: 44 gal

Whiteboards · Question 8 of 9

A plan is drawn at 1 in : 4 ft; a room is 2 in by 2.5 in on it. 1 ft = 0.305 m.

Find the room's area in square meters.

Answer it.

Whiteboards · Question 8 of 9

A plan is drawn at 1 in : 4 ft; a room is 2 in by 2.5 in on it. 1 ft = 0.305 m.

Find the room's area in square meters.

$2 \times 4 = 8$ ft $2.5 \times 4 = 10$ ft

$8 \times 0.305 = 2.44$ $10 \times 0.305 = 3.05$ $2.44 \times 3.05 = 7.442$

Answer: About 7.44 m²

Whiteboards · Question 9 of 9

1 mi = 1.61 km. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h.

Which car gets there first, and by how long? Explain.

Write your answer in sentences.

Whiteboards · Question 9 of 9

1 mi = 1.61 km. A 300-km drive. Car A averages 50 mi/h; car B averages 75 km/h.

Which car gets there first, and by how long? Explain.

$$50 \times 1.61 = 80.5 \text{ km/h}$$

$$300 \div 80.5 \approx 3.73 \text{ h} \qquad 300 \div 75 = 4 \text{ h}$$

Answer: Car A: 80.5 km/h against 75, so about 3.73 h against 4 h — about 16 minutes sooner.

Independent Set

1. A car gets 32 miles per gallon and its tank holds 14 gal. How far does it go on a tank?

2. That car makes a 1,000-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

3. A recipe for 4 servings uses 1¼ cups of rice. How much rice for 10 servings?

4. 1 km = 0.62 mi. A train does 120 km/h; a bus does 70 mi/h. Which is faster? Show the speeds in one unit.

5. Juice is mixed orange : mango : water = 3 : 2 : 5. A batch uses 9 cups of orange. How many cups is the whole batch?

6. A plan is drawn at 1 in : 6 ft; a room is 2 in by 3.5 in on it. Find the room's real area.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Compare rates: word problems — VVS

Multi-step problems with customary or metric unit conversions — UCT